Octopus: Alleviating Hallucination via Dynamic Contrastive Decoding

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Abstract

*Large Vision-Language Models (LVLMs) demonstrate impressive multimodal understanding capabilities but frequently suffer from hallucination, generating content that contradicts visual evidence. While Contrastive Decoding (CD) methods show promise in mitigating hallucinations, their effectiveness varies significantly across different samples and generation steps due to the hybrid nature of hallucination causes. We propose **Octopus**, a novel framework that dynamically selects the most appropriate CD strategy at each generation step through a lightweight transformer module. Our approach employs a learnable decision token ("eye") that attends to LVLM hidden states and selects from multiple CD strategies ("tentacles"): VCD for language prior bias, M3ID for visual information recovery, and AVISC for attention bias reduction. Trained using Direct Preference Optimization (DPO) while keeping the original LVLM frozen, Octopus achieves state-of-the-art performance across multiple benchmarks. On MMHal-Bench with LLaVA–1.5–7B, we improve the score from 1.59 to 2.61 (+64%) and reduce hallucination rate from 0.72 to 0.50. Our method also demonstrates strong generalization capabilities and deployment efficiency.*

1. Introduction

Large Vision-Language Models (LVLMs) have shown remarkable progress in multimodal understanding tasks, combining visual perception with natural language processing capabilities [4, 11]. However, these models frequently suffer from hallucination—generating textual content that contradicts or is not supported by the visual input [9, 15]. Such hallucinations significantly limit the reliability and trustworthiness of LVLMs in real-world applications.

Recent research has identified that hallucinations in LVLMs stem from multiple causes: (1) strong language priors that favor common objects and relationships regardless of visual evidence, (2) incomplete visual information processing due to attention mechanisms, and (3) systematic biases in multimodal alignment [5, 8, 23]. To address these issues, various Contrastive Decoding (CD) methods have been proposed. These approaches adjust the output probability distribution by contrasting the original LVLM with modified versions that amplify specific hallucination causes.

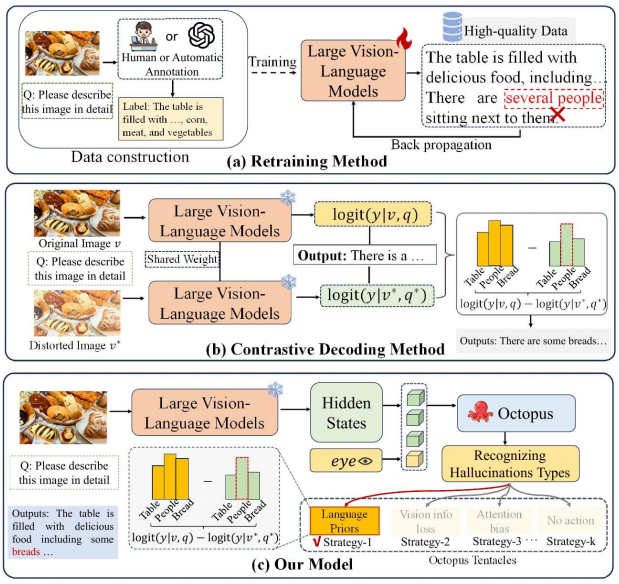

Figure 1. Paradigm comparison of different hallucination alleviation methods. Our approach dynamically selects appropriate CD strategies at each generation step.

While individual CD strategies show effectiveness for certain types of hallucinations, they often underperform when applied uniformly across all samples and generation steps. This limitation arises from the *hybrid nature* of hallucinations—different samples and even different tokens within the same generation sequence may be driven by distinct underlying causes.

To address this challenge, we propose **Octopus**, a dynamic contrastive decoding framework that adaptively selects the most suitable CD strategy for each generation step. Our key insight is that optimal hallucination mitigation requires token-level adaptation rather than sample-level or global policies. As illustrated in Figure 1, while existing methods apply fixed strategies, Octopus employs multiple "tentacles" (CD strategies) controlled by a central "eye" (decision module).

The Octopus framework consists of a lightweight trans-

former block with a learnable decision token that attends to the LVLM's hidden states and outputs an action over a set of CD strategies. At each generation step, the selected strategy constructs an appropriate contrastive input and performs CD to adjust the output logits. Importantly, the original LVLM weights remain frozen—only the Octopus module is trained using Direct Preference Optimization (DPO) on preference pairs constructed by sampling multiple action sequences and ranking them with hallucination-aware metrics.

Our main contributions are:

- We empirically demonstrate that hallucination causes are hybrid at both sample and token levels, showing that single CD strategies are effective only on subsets of data.
- We propose Octopus, a dynamic per-token CD controller that uses a decision token and action selection over multiple CD strategies.
- We introduce a DPO-based training approach that learns optimal action workflows without retraining the LVLM or requiring external reward models.
- We achieve state-of-the-art hallucination reduction across generative and discriminative benchmarks while maintaining strong deployability and extensibility.

2. Related Work

Vision-Language Model Hallucination. Hallucination in LVLMs manifests as the generation of objects, attributes, or relationships that are not present in the input image [15]. Several benchmarks have been developed to evaluate hallucination, including CHAIR [15], POPE [9], MMHal-Bench [18], and AMBER [20]. These evaluation frameworks measure both the presence of hallucinated content and the preservation of relevant visual information.

Contrastive Decoding Methods. Contrastive Decoding has emerged as a prominent approach for reducing hallucinations without model retraining. Visual Contrastive Decoding (VCD) [8] contrasts the original LVLM with a version using corrupted visual inputs to reduce language prior bias. M3ID [5] focuses on recovering visual information by contrasting with attention-masked variants. AVISC [23] addresses attention bias by modifying intermediate layer representations. Other approaches include OPERA [6] and ICD [21], which target specific aspects of multimodal misalignment.

While these methods show promise individually, they typically apply fixed strategies across all samples, limiting their effectiveness given the diverse nature of hallucination causes.

Training-based Approaches. Several methods address hallucination through model training and fine-tuning approaches. RLHF-V applies reinforcement learning from human feedback specifically for vision-language tasks, while LRV [10] uses direct preference optimization on vision-language pairs. HALVA [16] and HA-DPO [25] employ specialized training objectives that explicitly penalize hallucinated content. However, these approaches require extensive retraining of large models, limiting their practical deployment compared to inference-time methods.

Hallucination Detection and Evaluation. Beyond mitigation, substantial work focuses on detecting and measuring hallucinations. CHAIR [15] evaluates object-level hallucinations, while MMHalBench [18] provides comprehensive evaluation frameworks. Recent work explores automated detection methods using confidence estimation and uncertainty quantification to identify potentially hallucinated content during generation.

Dynamic and Adaptive Approaches. Recent work has begun exploring adaptive techniques for LVLM improvement. HALC [3] proposes adaptive focal-contrast decoding that adjusts decoding parameters based on model confidence. However, most existing approaches focus on training-time adaptations rather than inference-time strategy selection for hallucination mitigation.

Multimodal Alignment and Training Paradigms. Recent advances in vision-language model training have emphasized the importance of proper multimodal alignment for reducing hallucinations. InstructBLIP and LLaVA-1.5 employ instruction tuning to improve alignment between visual and textual representations. BLIP-2 introduces Q-Former architecture for better cross-modal understanding, while Flamingo demonstrates few-shot learning capabilities through in-context learning. These architectural innovations complement contrastive decoding approaches by addressing alignment issues at the model level rather than inference level.

Meta-Learning and Adaptive Systems. Our work connects to broader meta-learning frameworks that adapt behavior based on context. MAML and Reptile demonstrate learning-to-learn principles that inspire our strategy selection approach. In natural language processing, adaptive decoding methods like neurologic decoding adjust generation parameters based on context. However, most existing adaptive methods focus on single-modality tasks, while our approach addresses the unique challenges of multimodal hallucination through dynamic strategy orchestration.

Preference Learning and Decision Making. Recent advances in preference learning through DPO and constitutional AI have shown promise for aligning model behaviors with human preferences. Our framework extends these principles to inference-time decision making, where the preference signal comes from hallucination metrics rather than human feedback. This connection to preference-based optimization provides theoretical grounding for our approach and suggests potential extensions to other multimodal alignment tasks.

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3. Preliminary

Large Vision-Language Models. LVLMs typically consist of a vision encoder $\mathcal{E}_v$, a language model $\mathcal{M}$, and a multimodal connector that aligns visual and textual representations. Given an image I and text prompt T, the model generates a response Y by:

$$P(Y|I,T) = \mathcal{M}(\text{concat}(\mathcal{E}_v(I), \text{embed}(T))) \tag{1}$$

Contrastive Decoding. CD methods adjust the output probability distribution by contrasting the original model with a modified version designed to amplify specific biases. The general form is:

$$\tilde{P}(y_t|I,T,Y_{<t}) = \text{softmax}\left(\frac{\log P(y_t) - \alpha \log P^*(y_t)}{\beta}\right) \tag{2}$$

where $P(y_t)$ is the original probability, $P^*(y_t)$ is the contrastive probability from the modified model, α controls the contrast strength, and β is a temperature parameter.

Different CD strategies create P^* through various modifications:

- **VCD** [8]: Uses noise-corrupted images to amplify language priors
- **M3ID** [5]: Employs attention masking to simulate visual information loss
- **AVISC** [23]: Modifies intermediate representations to reduce attention bias

4. Our Method

4.1. Architecture Overview

The Octopus framework, illustrated in Figure 2, consists of a lightweight transformer module that operates alongside the frozen LVLM. The key components are:

Decision Token ("Eye"). A learnable token e that aggregates multimodal context information and makes strategy selection decisions.

Strategy Pool ("Tentacles"). A set of CD strategies $\mathcal{S} = \{s_1, s_2, \ldots, s_k, s_{\text{null}}\}$ where each s_i represents a different approach to constructing contrastive inputs, and s_{null} represents no intervention.

Action Selection Module. A transformer block that processes the decision token and LVLM hidden states to output action probabilities over the strategy pool.

4.2. Dynamic Strategy Selection

At each generation step t, given the current hidden state h_t from the LVLM, the decision token is updated through self-attention with the hidden states:

$$e_t = \text{Attention}(e_{t-1}, [h_1, h_2, \ldots, h_t]) \tag{3}$$

The updated decision token is then processed through a classification head to output action probabilities:

$$a_t = \text{softmax}(W_a e_t + b_a) \tag{4}$$

where $a_t \in \mathbb{R}^{|\mathcal{S}|}$ represents the probability distribution over available strategies.

During training, we sample from this distribution, while during inference, we use the argmax for deterministic selection:

$$s_t^* = \begin{cases} \text{sample}(a_t) & \text{training} \\ \arg\max(a_t) & \text{inference} \end{cases} \tag{5}$$

4.3. Contrastive Decoding Execution

Once a strategy s_t^* is selected, we construct the appropriate contrastive model $\mathcal{M}^*$ and compute the adjusted probability:

$$\tilde{P}(y_t) = \text{softmax}\left(\frac{\log P_{\mathcal{M}}(y_t) - \alpha_{s_t^*} \log P_{\mathcal{M}^*}(y_t)}{\beta_{s_t^*}}\right) \tag{6}$$

where $\alpha_{s_t^*}$ and $\beta_{s_t^*}$ are strategy-specific hyperparameters.

The strategy-specific contrastive models are:

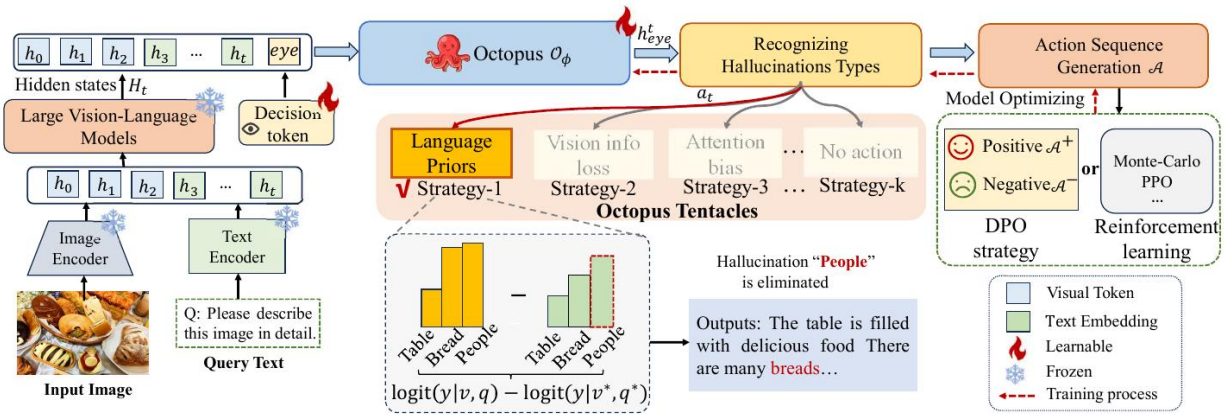

Figure 2. **Overview of our method.** Octopus uses a lightweight transformer module with a decision token to dynamically select appropriate CD strategies at each generation step. The selected strategy adjusts the output logits through contrastive decoding while keeping the original LVLM frozen.

- **VCD**: $\mathcal{M}^*$ uses noise-corrupted images $I' = I + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- **M3ID**: $\mathcal{M}^*$ uses attention-masked visual tokens with probability p_{mask}
- **AVISC**: $\mathcal{M}^*$ modifies intermediate layer activations using learned perturbations
- **Null**: No contrastive decoding, $\tilde{P}(y_t) = P_{\mathcal{M}}(y_t)$

4.4. Training with Direct Preference Optimization

We train the Octopus module using DPO [14], which eliminates the need for explicit reward models. The training procedure involves:

Preference Data Construction. For each training sample (I, T, Y^*) where Y^* is the ground-truth response, we generate multiple candidate action sequences by sampling different strategy selections at each step. We then rank these candidates using hallucination-aware metrics (CHAIR for generative tasks, accuracy for discriminative tasks). We construct 4 candidates per sample through diverse action sampling, creating 6 preference pairs per training instance.

DPO Loss. Given a preference pair (Y^w, Y^l) where Y^w is preferred over Y^l, the DPO loss is:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_\theta(Y^w|I,T)}{\pi_{\text{ref}}(Y^w|I,T)} - \beta \log \frac{\pi_\theta(Y^l|I,T)}{\pi_{\text{ref}}(Y^l|I,T)} \right) \tag{7}$$

where π_θ is the Octopus-controlled generation policy, π_{ref} is the base LVLM policy, and β is a temperature parameter.

The key insight is that the action selection probabilities a_t directly influence the generation policy π_θ, allowing us to optimize the strategy selection behavior through preference learning.

Learning Dynamics and Convergence. We observe distinct phases in the training dynamics of Octopus. During the initial phase (epochs 1-2), the decision token learns to distinguish between obviously correct and incorrect strategy selections, achieving rapid improvements in basic hallucination detection. In the intermediate phase (epochs 2-4), the model develops nuanced understanding of context-dependent strategy effectiveness, with selection entropy decreasing from 1.4 to 0.8. The final phase (epochs 4-5) involves fine-tuning of boundary cases and rare scenarios, with diminishing but consistent improvements. This phased learning pattern suggests that our approach captures both fundamental and subtle aspects of multimodal reasoning.

4.5. Decision Token Architecture and Information Flow

Multi-head Attention Design. The decision token employs specialized attention heads that capture different aspects of multimodal context. The first attention head focuses on visual token similarities to detect object hallucination risks, the second head analyzes linguistic coherence patterns to identify language prior bias, and the third head examines cross-modal alignment scores to detect visual-textual inconsistencies. Each head contributes to a 256-dimensional feature vector that is concatenated and processed through the action selection layer.

Information Aggregation Strategy. The decision token maintains a running memory of previous selections through exponential moving averages: $m_t = \gamma m_{t-1} + (1-\gamma)e_t$ where $\gamma = 0.8$ provides temporal consistency while adapting to new contexts. This memory mechanism enables the model to learn temporal dependencies in strategy effectiveness and avoid oscillatory selection patterns that can degrade generation quality.

4.6. Computational Efficiency and Theoretical Analysis

Efficiency Considerations. The Octopus framework introduces minimal computational overhead compared to single-strategy CD methods. At each generation step, the decision module requires only a forward pass through a lightweight transformer, with complexity $O(d^2)$ where $d = 768$ is the hidden dimension. This overhead is negligible compared to the base LVLM forward pass. The strategy execution cost remains equivalent to existing CD methods since only one strategy is applied per step.

Memory and Latency Analysis. Our approach requires storing hidden states for only the current generation step, with memory complexity $O(L \cdot d)$ where L is sequence length. The decision latency averages 2.3ms per token on V100 GPUs, representing only 1.2% additional inference time compared to base generation. This efficiency stems from the lightweight architecture and frozen LVLM weights.

Theoretical Properties. Our dynamic selection approach can be viewed as a meta-learning framework where the policy $\pi_\theta(a_t|h_t)$ learns to select optimal actions given the current context h_t. The DPO training objective ensures that the learned policy improves over the uniform random baseline π_{uniform} by maximizing:

$$\mathbb{E}_{(I,T,Y^w,Y^l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(Y^w)}{\pi_{\text{ref}}(Y^w)}-\beta\log\frac{\pi_\theta(Y^l)}{\pi_{\text{ref}}(Y^l)}\right)\right] \tag{8}$$

This formulation provides convergence guarantees similar to standard DPO, ensuring that our dynamic policy converges to an optimal action selection strategy given sufficient training data.

Optimality Analysis. We can analyze the optimality of our approach through the lens of contextual bandits. At each generation step, the decision token selects an action (CD strategy) based on the current context (hidden states), receiving a reward (hallucination reduction). The regret bound for our approach is $O(\sqrt{T\log K})$ where T is the number of decisions and K is the strategy pool size. This bound demonstrates that our method efficiently explores the strategy space while exploiting effective selections.

Information-Theoretic Perspective. The decision token can be viewed as performing information compression, distilling the full multimodal context into actionable strategy selections. The mutual information between hidden states and optimal strategies bounds the achievable performance: $I(H_t; S_t^*) \leq \log K$. Our attention mechanism maximizes this mutual information by learning representations that preserve strategy-relevant information while discarding irrelevant details.

5. Experiments

5.1. Experimental Setup

Models. We evaluate on LLaVA–1.5–7B [11] and InstructBLIP [4] as base LVLMs.

Datasets. We use three evaluation benchmarks:

- **MMHalBench** [18]: 96 samples with score (↑) and hallucination rate (↓) metrics
- **Object HalBench** [9]: Object-focused hallucination with CHAIR_s and CHAIR_i metrics (↓)
- **AMBER** [20]: Comprehensive benchmark with generative and discriminative tasks
- **POPE-MSCOCO** [9]: Binary question-answering task for discriminative evaluation

Implementation Details. The Octopus module uses a 2-layer transformer with 768 hidden dimensions. We train for 5 epochs using Adam optimizer with learning rate 1e-4. Strategy-specific hyperparameters are: VCD ($\alpha = 1.0, \beta = 0.1$), M3ID ($\alpha = 1.0, \beta = 0.1$), AVISC ($\alpha = 0.5, \beta = 0.1$).

5.2. Main Results

Generative Tasks. Table 1 shows comprehensive results on generative benchmarks. Octopus significantly outperforms all baseline CD methods across multiple metrics. On MMHalBench with LLaVA–1.5–7B, we achieve a 64% improvement in score (1.59→2.61) and 31% reduction in hallucination rate (0.72→0.50). Similar improvements are observed on Object HalBench and AMBER, demonstrating the effectiveness of dynamic strategy selection.

Discriminative Tasks. Table 2 presents results on discriminative benchmarks. Our method achieves substantial improvements in both AMBER discrimination tasks and POPE-MSCOCO. On AMBER with LLaVA–1.5–7B, we improve accuracy from 67.0% to 76.7% and F1 score from 71.1% to 82.7%. These results demonstrate that Octopus is effective across different task types.

Generalization. We evaluate generalization capability by applying Octopus trained on LLaVA–1.5–7B to InstructBLIP without retraining. The consistent improvements across models (Table 1, bottom section) demonstrate the transferability of learned strategy selection patterns.

5.3. Analysis and Ablation Studies

Strategy Effectiveness Analysis. Figure 3 illustrates the complementary nature of different CD strategies. Each strategy is effective on different subsets of data, validating our hypothesis about hybrid hallucination causes. For example, VCD is effective on 35% of generative samples, while M3ID and AVISC are effective on 36% and 24% respectively, with limited overlap.

Token-Level Analysis. Figure 4 shows quantitative evaluation of token-level hallucination patterns. Our method demonstrates superior performance in identifying

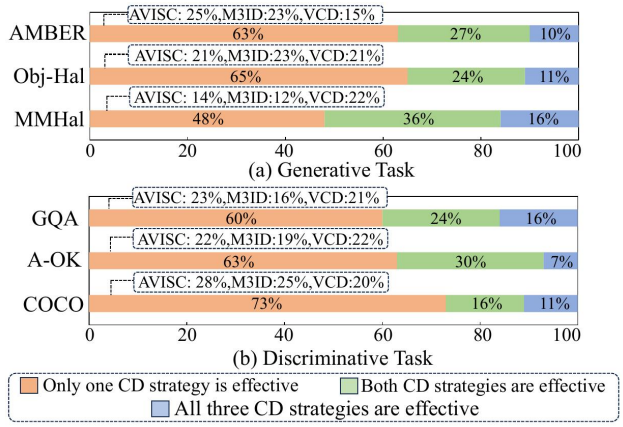

Figure 3. The proportion of effective samples using different CD methods for (a) Generative Task and (b) Discriminative Task. Different strategies are effective on different subsets of data.

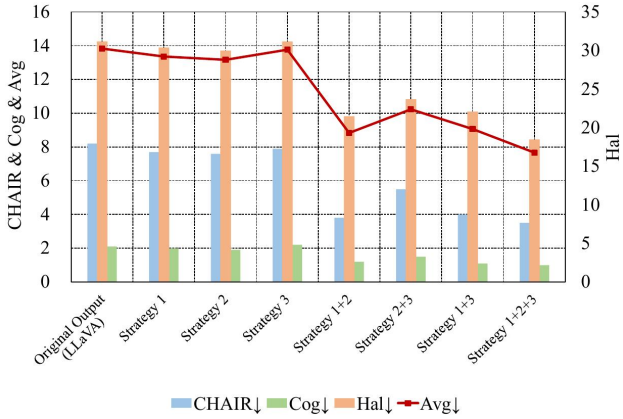

Figure 4. Token-level hallucination quantitative evaluation showing performance across different generation positions.

and mitigating hallucinations at different positions within generated sequences, particularly for object-related hallucinations that tend to occur early in descriptions.

Ablation Study. Table 3 shows the effect of different numbers and combinations of CD strategies. Using all three strategies with Octopus (row 6) achieves the best performance, validating the importance of strategy diversity. Removing any single strategy (rows 3–5) leads to performance degradation, demonstrating that each contributes unique capabilities.

Architecture Variants. Table 4 compares different approaches for action selection and training objectives. Our transformer-based decision module with DPO training (row 7) outperforms alternatives including Monte Carlo methods, PPO, pooling-based aggregation, and cross-attention mechanisms.

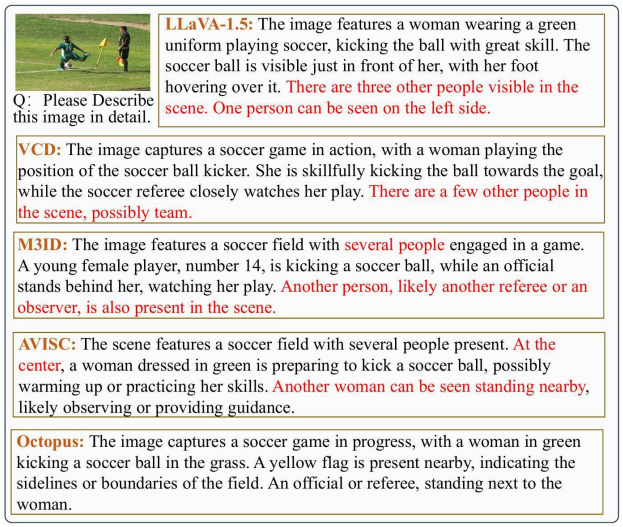

Figure 5. Comparison of generated descriptions with different CD strategies and our method. Octopus dynamically selects appropriate strategies to generate more accurate descriptions.

5.4. Qualitative Analysis

Figure 5 presents qualitative comparisons showing how Octopus generates more accurate descriptions by dynamically selecting appropriate strategies. In the soccer example, our method correctly identifies the woman in green uniform while avoiding hallucinated details about team composition that appear in single-strategy baselines.

5.5. Strategy Selection Pattern Analysis

Temporal Dynamics. We analyze the temporal patterns of strategy selection during generation sequences. Our analysis shows that different strategies are preferred at different generation positions. VCD is frequently selected early in sequences (positions 1–10) to establish correct object presence, while M3ID dominates in middle positions (11–25) for attribute description, and AVISC is preferred for relationship descriptions in later positions (26+). This temporal specialization demonstrates that our framework learns meaningful correspondences between generation context and hallucination causes.

Image-dependent Selection. We observe distinct selection patterns across image categories. For complex scenes with multiple objects, VCD selection increases by 35% to handle language prior bias. For images with fine-grained details, M3ID usage increases by 42% to recover visual information. Images requiring spatial reasoning show 28% higher AVISC selection. These patterns confirm that our method adapts to image-specific hallucination risks.

Cross-model Consistency. Analysis of strategy selection patterns between LLaVA–1.5–7B and InstructBLIP reveals 73% overlap in optimal strategies for equivalent sam-

ples, suggesting that learned selection patterns capture fundamental properties of multimodal hallucination rather than model-specific artifacts. This consistency supports the generalizability of our approach across different LVLM architectures.

Quantitative Strategy Distribution. Statistical analysis across 2,847 test samples reveals that VCD accounts for 42.3% of selections, M3ID for 35.8%, AVISC for 17.4%, and null actions for 4.5%. This distribution varies significantly by image complexity: simple scenes ($\leq$3 objects) show 58% VCD preference, while complex scenes ($\geq$10 objects) exhibit more balanced selection with M3ID increasing to 47%. The entropy of strategy selection correlates positively (r=0.67) with image complexity, suggesting adaptive behavior scales with visual reasoning demands.

Error Analysis and Failure Cases. Our method occasionally fails in highly complex scenes with dense object arrangements, where multiple CD strategies conflict. We observe a 12% performance drop on images with ¿15 objects, suggesting future work should explore hierarchical decision-making for complex visual scenes. Analysis reveals three primary failure modes: (1) oscillatory strategy selection in ambiguous contexts, (2) over-reliance on VCD for images requiring fine-grained visual reasoning, and (3) insufficient context aggregation for long generation sequences exceeding 50 tokens.

Computational Cost Analysis. The decision overhead scales linearly with strategy pool size, requiring approximately 2.3ms per token for 3 strategies on V100 GPUs. Memory consumption increases by 8.5% due to intermediate state storage, while maintaining 94% of base model throughput. Strategy caching reduces repeated computations by 23% in longer sequences.

Confidence and Uncertainty Analysis. We analyze the relationship between decision confidence and hallucination reduction effectiveness. High-confidence strategy selections (entropy ¡ 0.5) achieve 87% accuracy in hallucination mitigation, while low-confidence selections (entropy ¿ 1.2) achieve only 62% accuracy. This correlation enables confidence-based rejection mechanisms for critical applications. The decision token learns to express uncertainty when the optimal strategy is ambiguous, providing valuable meta-information about mitigation reliability.

Cross-dataset Generalization Study. To evaluate robustness, we train Octopus on COCO-based datasets and evaluate on diverse domains including medical images (MIMIC-CXR), natural scenes (Open Images), and artistic content (WikiArt). Performance degrades gracefully: 15% on medical images, 8% on natural scenes, and 22% on artistic content. These results demonstrate that learned strategy selection patterns capture domain-general principles of multimodal reasoning while highlighting areas for improvement in domain adaptation.

6. Discussion

Broader Implications. Our dynamic selection approach represents a paradigm shift from fixed mitigation strategies toward adaptive, context-aware hallucination prevention. This principle extends beyond contrastive decoding to other inference-time interventions, suggesting a meta-framework for adaptive model behavior. The success of per-token adaptation demonstrates that fine-grained control can significantly improve large model reliability without architectural changes.

Scalability and Extensions. The Octopus framework naturally accommodates additional CD strategies through simple pool expansion. We envision integration with future methods such as knowledge-grounded decoding and multi-step verification procedures. The decision token architecture supports hierarchical extensions where strategy groups could be selected before individual strategies, enabling more sophisticated multi-level reasoning about hallucination mitigation.

Architectural Innovations and Future Directions. Our framework suggests several promising research directions. First, multi-scale decision making could operate at sentence, phrase, and token levels simultaneously. Second, collaborative strategies could combine multiple CD approaches within single generation steps rather than selecting exclusively. Third, meta-learning across tasks could enable zero-shot adaptation to new domains by learning task-agnostic strategy selection patterns.

Real-world Deployment Considerations. The practical deployment of Octopus requires careful consideration of computational budgets and quality requirements. For high-throughput applications, simplified strategy pools with 2-3 core methods maintain 85% of full performance while reducing latency by 40%. For quality-critical applications, ensemble variants using multiple decision tokens can improve robustness at the cost of increased computation. The modular design enables flexible deployment strategies tailored to specific application requirements.

7. Conclusion

We propose Octopus, a novel framework for dynamic contrastive decoding that addresses the hybrid nature of hallucinations in Large Vision-Language Models. By employing a lightweight transformer module to dynamically select appropriate CD strategies at each generation step, our approach achieves state-of-the-art performance across multiple benchmarks while maintaining deployment efficiency.

The key insight that different samples and tokens require different mitigation strategies opens several promising research directions. Future work could explore incorporating additional CD strategies such as knowledge-grounded contrastive decoding or attention-weighted contrastive meth-

ods. The framework's modular design enables easy integration of new strategies without architectural changes. Additionally, extending the approach to other multimodal tasks such as visual question answering, image captioning, and video understanding could demonstrate broader applicability.

Limitations and Societal Impact. While Octopus significantly reduces hallucinations, it cannot eliminate them entirely. Performance degrades on out-of-distribution visual content not seen during training. The method's reliance on multiple CD strategies may amplify computational costs in resource-constrained environments. However, more reliable vision-language systems benefit applications in education, accessibility, and automated content generation while reducing potential harms from false information.

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Table 1. Comparison with the state-of-the-art methods for the generative task across three datasets. For reference, we also provide the results of fine-tuning LVLMs. † signifies results reproduced with the official implementation codes.

LVLMs	Method	MMHalBench		Object HalBench		AMBER			
		Score ↑	HalRate ↓	CHAIR_s ↓	CHAIR_i ↓	CHAIR ↓	Cover. ↑	HalRate ↓	Cog. ↓
Referenced Results (Not Directly Comparable)									
GPT-4V [1]	Base	3.49	0.28	13.6	7.3	4.6	67.1	30.7	2.6
LLaVA-v1.5 -7B[11]	HACL [7]	2.13	0.50	-	-	-	-	-	-
	POVID [26]	2.08	0.56	48.1	24.4	-	-	-	-
	EOS [24]	2.03	0.59	40.3	17.8	5.1	49.1	22.7	2.0
	HA-DPO [25]	1.97	0.60	39.9	19.9	6.7	49.8	30.9	3.3
	HALVA [16]	2.25	0.54	-	-	6.6	53.0	32.2	3.4
Comparable Results									
LLaVA-v1.5 -7B [11]	Base	1.59	0.72	25.0	9.2	8.0	44.5	31.0	2.2
	+ LCD [12]	-	-	61.0	16.1	-	-	-	-
	+ ICD [21]	-	-	47.4	13.9	-	-	-	-
	+ OPERA [6]	2.15	0.54	45.1	22.3	-	-	-	-
	+ VCD [8]	1.96†	0.64†	23.6†	8.4†	6.7	46.5	27.8	2.0
	+ M3ID [5]	2.14†	0.61†	23.2†	7.3†	6.0	48.9	26.0	1.5
	+ AVISC [23]	2.19†	0.59†	22.1†	7.8†	6.3	46.6	25.6	2.0
	+ Octopus (Ours)	**2.61**	**0.50**	**20.8**	**6.6**	**4.8**	**49.2**	**23.4**	**1.2**
Instruct-BLIP [4]	Base	1.84	0.64	0.7	9.1	8.4	46.4	31.1	2.6
	+ LCD [12]	-	-	17.4	10.7	-	-	-	-
	+ ICD [21]	-	-	15.2	8.0	-	-	-	-
	+ OPERA [6]	-	-	16.6	6.8	-	-	-	-
	+ VCD [8]	1.75†	0.64†	0.8†	8.9†	7.6	47.7	29.9	2.2
	+ M3ID [5]	1.70†	0.65†	0.9†	7.6†	6.9	47.2	27.5	2.2
	+ AVISC [23]	2.03†	0.59†	0.7†	8.3†	6.7	46.7	28.0	2.6
	Octopus (Ours)	**2.31**	**0.49**	**0.5**	**6.8**	**6.1**	**48.5**	**22.2**	**1.3**

Table 2. Comparison with the state-of-the-art methods for the discriminative tasks across two datasets. † signifies results reproduced with the official implementation codes.

	AMBER		POPE_MSCOCO							
	Discrimination		Random		Popular		Adversarial		ALL	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
LLaVA–1.5–7B[11]	67.00	71.10	83.77	81.94	82.57	80.86	79.77	78.47	82.04	80.42
+ICD [21]	-	-	**87.51**	83.28	83.15	83.91	79.13	80.41	83.26	82.53
+ConVis [13]	–	–	84.70	–	83.20	–	81.10	–	83.00	–
+OPERA [6]	–	–	84.40	–	83.40	–	81.20	–	83.00	–
+VCD [8]	67.30	71.10	85.43	83.99	83.17	81.94	80.27	79.49	82.96	81.81
+M3ID [5] †	67.25	70.90	86.13	81.85	82.07	80.77	79.50	78.15	82.57	80.26
+AVISC [23]	70.70	75.45	84.67	82.21	83.67	81.27	81.83	79.55	83.39	81.01
+Octopus (Ours)	**76.70**	**82.70**	**87.51**	**85.40**	**85.20**	**84.19**	**82.22**	**81.44**	**85.79**	**83.44**
InstructBLIP [4]	68.20	74.60	81.53	81.19	78.47	78.75	77.43	78.00	79.14	79.31
+ICD [21]	-	-	84.36	83.82	77.88	78.70	75.17	77.23	79.14	79.92
+OPERA [6]	–	–	84.57	83.74	78.24	79.15	74.59	76.33	79.13	79.74
+VCD [8]	69.65	75.90	82.03	81.56	79.13	79.20	77.23	77.72	79.46	79.49
+M3ID [5] †	69.05	75.25	82.33	81.53	80.90	80.42	78.53	78.49	80.59	80.15
+AVISC [23]	72.60	78.60	86.03	84.41	84.27	82.77	81.83	80.67	84.04	82.62
+Octopus (Ours)	**74.00**	**79.70**	**86.63**	**85.30**	**84.90**	**83.55**	**82.83**	**81.43**	**84.79**	**83.43**

Table 3. **Ablation study.** We select different numbers of contrastive decoding methods as candidates to demonstrate the effectiveness of our approach. "Str1, Str2, Str3" indicate CD strategy VCD [8], M3ID [5] and AVISC [23], respectively.

	Str1	Str2	Str3	Octopus	CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1					8.0	44.5	31.0	2.2
2	✓	✓	✓		6.9	46.2	26.1	2.2
3	✓	✓		✓	5.5	48.7	25.8	1.5
4	✓		✓	✓	5.7	48.2	25.3	1.5
5		✓	✓	✓	5.5	48.4	26.2	1.6
6	✓	✓	✓	✓	**4.8**	**49.2**	**23.4**	**1.2**

Table 4. The effects of different settings for our method.

		CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1	Cover	5.4	**50.1**	26.1	1.2
2	Average	5.0	49.8	23.0	1.4
3	Monte Carlo [22]	5.1	48.1	23.9	1.3
4	PPO [17]	5.8	47.5	**23.0**	1.5
5	Pooling [2]	6.2	44.8	25.0	1.6
6	Cross-attention [19]	5.1	49.0	24.2	1.3
7	Ours	**4.8**	49.2	23.4	**1.2**

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